



DPUMERedlab

Interface for external measurement instruments from the Meilhaus Redlab family. Optimized for model Redlab 1208 FS.

1. Inputs:

AIn[01..02] Value of the analog output of the measurement in-strument ranging from 0 to 1. The real voltage (V) depends on the used instrument. With the Redlab 1208 FS it is between 0 V and 4 V.

2. I/Os:

Port0.Bit[01..08] Bit 1 - 8 of the digital port A
Port1.Bit[01..08] Bit 1 - 8 of the digital port B

3. Static parameters:

BoardNumber Number of the measurement instrument as shown by InstaCAL. (default: 0 = first connected instrument)

AInChannels Number of analog inputs to poll [0..8] (default: 4)
The more inputs are polled, the lower is the achievable sample rate. Using Redlab 1208 FS around 500 Hz can be reached with four inputs active.

AInVoltage Measurement range for analog inputs in V. This range is symmetrical from *-AInVoltage* to *AInVoltage*.
Redlab 1208 FS supports a range of 1 V to 20 V.

AOutChannels Number of active analog outputs [0..2] (default: 0). The more outputs are activated, the more calculation time is needed by DPUMERedlab.

Port[0..1].Mode Mode of the digital I/O port. Each port is 8 bit which signals are available as variables *Porti.Bitxx*.
0 (default): do not use
1: use port as input
2: use port as output

4. Outputs:

AOut[01..08] Voltage (V) of the analog input of the measurement instrument.

5. Functional principle:

DPUMERedlab supplies an interface for the USB measurement instruments of the Redlab series from Meilhaus. Names of the inputs and outputs are done by the cards point of view.

Several Redlab devices may be connected to a PC at the same time. Before using the device has to be configured and calibrated by the enclosed software InstaCAL. During this process every device gets a



unique number (0,1,2,...). This number has to be set as parameter *BoardNumber* (default: 0) for the DPU.

Single functions of the Redlab devices (analog inputs, analog outputs und digital I/O ports) can be activated separately (using parameters *AInChannels*, *AOutChannels* and *Porti.Mode*). It is recommended to activate only necessary functions to reduce the calculation time of DPUMERedlab and get the highest possible sample rate for the analog inputs. If only a part of the inputs or outputs are set active, the channels with the lowest numbers will be used. For example: If *AInChannels* is set to 2 then only analog inputs CH0 and CH1 are polled.

Analog inputs are polled with the same frequency as is DPUMERedlab running. In case this exceeds the maximum data rate of the device, all analog inputs are set to 0 values and an error message about data loss will be shown on stop of the simulation.